Managing Third-party Risks in Supply Chain Data Security: The Differentially Private Newsvendor

Chen Du (陈督), Chua A. Geoffrey
Nanyang Business School, NTU

POMS-China 2023 at ZJU
July, 2023

Data Security in Supply Chain

- Proprietary data, with the help of advanced ML, can build informational advantages for retail firms: identify demand trends (Spiegel et al. 2013), make better inventory decisions (Qi et al. 2023).
- However, proprietary data may fall into third-parties (competitors, suppliers, and business partners), causing far-reaching consequences:

(a) competition from horizontal competitors

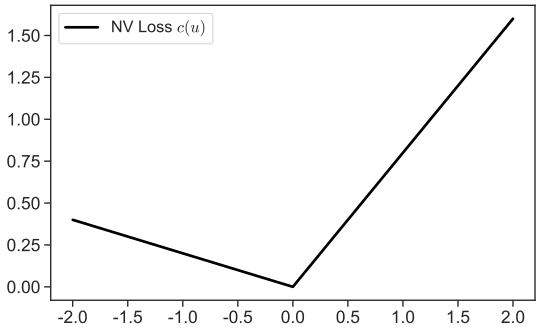
(b) disintermediation from vertical suppliers

(c) data leakage at business partners

Figure: severe consequences and various sources of data leakage

Data Leakage via Inference Attacks

- Security-unaware data-driven algorithms may disclose data **indirectly**, e.g.:
  - higher order quantity implies higher sales.
  - lower clearance price implies slow-moving.
- The data-driven newsvendor problem:
  - Binary demand $y \in \{0, 1\}$; critical ratio $r = 0.5$; order quantity q
  - Newsvendor cost $c(u) = ru^+ + (1-r)(-u)^+$, where underage level $u := y - q$
  - **SAA decision rule**: if ratio of 1 is strictly more than half, then order 1.



- Inference attack example:
  - A historical dataset:

Day	1	2	3	4	5
$y_{1:5}$	0	0	1	1	1

  - Then $\widehat{q}_6 = 1$. If $y_6 = 0$, then $\widehat{q}_7 = 0$.
  - Third-parties who observe $\widehat{q}_6, \widehat{q}_7$ can infer $y_6 = 0$, a **data leakage!**

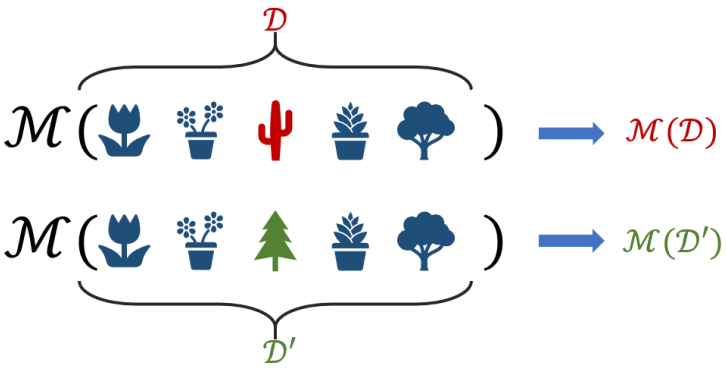
Differential Privacy: a unified and secure-by-design framework

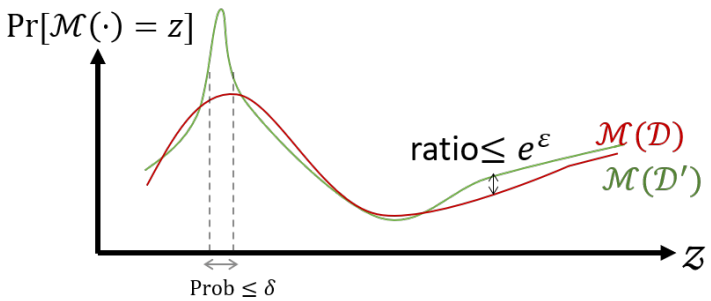
- A unified approach is in urgent need to prevent inference attacks.
- **Differential Privacy** (DP):

  For any given $\varepsilon \in \mathbb{R}_+, \delta \in [0,1]$, a randomized algorithm $\mathcal{M}$ is said to be (ε, δ)-differentially private, or (ε, δ)-DP in short, if

$$\Pr\left[\mathcal{M}(\mathcal{D}) = z\right] \leq e^{\varepsilon} \cdot \Pr\left[\mathcal{M}(\mathcal{D}') = z\right] + \delta, \quad \forall z \in \mathcal{Z}, \forall \mathcal{D} \sim \mathcal{D}',$$

  where $\mathcal{D} \sim \mathcal{D}'$ are neighboring datasets.





- **Why DP?** (1) Quantitative (2) Robust (3) data security standard in industry.
- **How to achieve DP?** by perturbing outputs of a data-driven algorithm.

Motivating Example Revisited

Binary demand $y \in \{0, 1\}$; critical ratio $r = 0.5$; historical dataset:

Day	1	2	3	4	5
$y_{1:5}$	0	0	1	1	1

Based on the historical dataset,

- Non-private decision by SAA: $\widehat{q}_6 = 1$
- $(\varepsilon, 0)$-differentially private decision:

$$\widehat{q}_6 = \begin{cases} 0, & \text{with prob. } \frac{1}{1+\exp(\varepsilon)}; \\ 1, & \text{with prob. } \frac{\exp(\varepsilon)}{1+\exp(\varepsilon)}. \end{cases}$$

Research Questions

- Of particular interest is to understand the **security-optimality trade-off**.

Research Questions:

1. How to design DP ordering algorithms for a data-driven contextual newsvendor? What is the cost of data security?
2. What are the drivers of the security cost? How to mitigate the cost?
3. What is the impact on upstream suppliers?

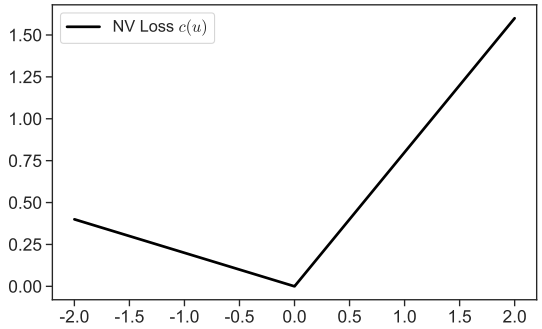
NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Related Literature

1. Data-Driven Newsvendor Problem:
   - Sampling case: Liyanage and Shanthikumar (2005), Levi et al. (2007), Chu et al. (2008), Ramamurthy et al. (2012), Levi et al. (2015), Cheung and Simchi-Levi (2019), Besbes and Mouchtaki (2021), Lin et al. (2022)
   - Contextual case: Huber et al. (2019), Ban and Rudin (2019), Oroojlooyjadid et al. (2020), Qi et al. (2022), Zhang et al. (2023), Qi et al. (2023), Besbes et al. (2023)
2. Differentially Private Algorithm Design:
   - Kifer et al. (2012), Bassily et al. (2014), Dwork et al. (2014), Vadhan (2017), Bassily et al. (2019), Feldman et al. (2020), Bassily et al. (2020), Asi et al. (2021), Bassily et al. (2021)
3. Growing literature on DP Operations Management:
   - **Personalized Pricing & Recommendation**: Chen et al. (2022a,b), Fu et al. (2022), Lei et al. (2023)
   - **Inventory Control**: Our work

The Differentially Private Newsvendor Problem (DP-NV)

- Newsvendor cost $c(u) = ru^+ + (1-r)(-u)^+, \forall u \in \mathbb{R}$.
- Demand follows $y = \langle \boldsymbol{\theta}^*, \boldsymbol{x} \rangle + \epsilon(\boldsymbol{x})$ with $\boldsymbol{\theta}^* \in \mathbb{R}^d$, where $\boldsymbol{x} \in \mathcal{X} \subseteq \mathbb{R}^d$ is the features, $y \in \mathcal{Y} \subseteq \mathbb{R}$, and $\epsilon(\boldsymbol{x})$ is an endogenous error term.
- The joint unknown distribution $(\boldsymbol{x}, y) \sim \mathbb{P}$



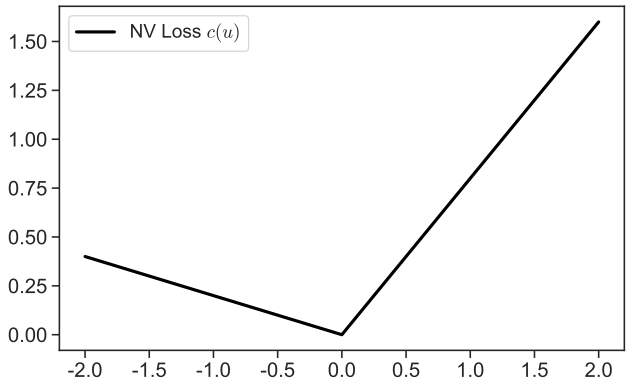
- Suppose the dataset to protect is $\mathcal{D} := \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n \sim \mathbb{P}^n$
- The (ε, δ)-differentially private newsvendor problem (DP-NV) is modelled as

$$\inf_{\pi \in \mathscr{F}_{\varepsilon,\delta}} \left\{ \sup_{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})} \mathcal{R}(\pi; \mathbb{P}) \right\},$$

  - where $\mathcal{R}(\pi; \mathbb{P}) := \mathbb{E}_{\mathcal{D},\pi}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}^\pi; \mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*; \mathbb{P})$ is the excess generalization risk of $\widehat{\boldsymbol{\theta}}^\pi$ given by policy π;
  - and $\mathcal{L}(\boldsymbol{\theta}; \mathbb{P}) := \mathbb{E}_{(\boldsymbol{x},y)\sim\mathbb{P}}\left[c(y - \langle \boldsymbol{\theta}, \boldsymbol{x} \rangle)\right]$ is the generalization risk of $\boldsymbol{\theta}$ under $\mathbb{P}$.

Convolution Smoothing

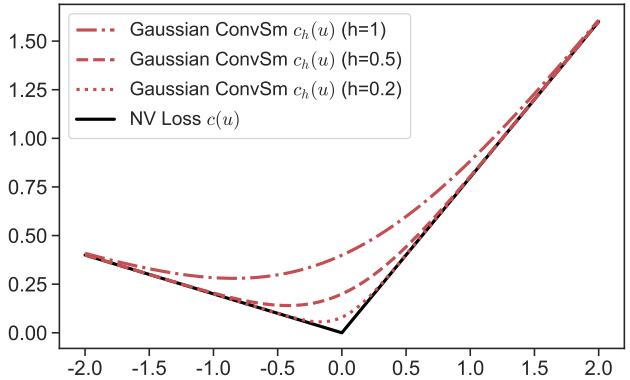
- **Technical Challenge**[1]: $c(\cdot)$ is nonsmooth
  - can leak much information
  - prevents optimization from being efficient



- **Convolution Smoothing**:

$$c_h(u) := \int_{-\infty}^{\infty} c(v) \cdot K_h(u - v)\, dv$$

where $K_h : \mathbb{R} \to \mathbb{R}_+$ is a kernel function parameterized by bandwidth $h > 0$.



[1] the work on addressing the technical challenge has been accepted at ICML 2023

DP Algorithm: Gradient Perturbation (GP)

- **Main idea**: Solve an empirical risk minimization problem on $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$,

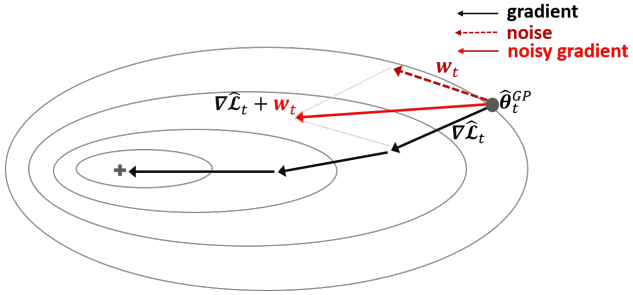
$$\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \widehat{\mathcal{L}}_h(\boldsymbol{\theta}; \mathcal{D}) := \frac{1}{n} \sum_{i=1}^n c_h(y_i - \langle \boldsymbol{\theta}, \boldsymbol{x}_i \rangle),$$

with fixed-step Gradient Descent, but perturb gradients in each iteration.

- **Key step**:
  1. draw Gaussian noise $\boldsymbol{w}_t \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$, and $\sigma^2 \asymp \ln(1/\delta)/\varepsilon^2$, update

     $$\widehat{\boldsymbol{\theta}}^{\mathrm{GP}}_{h,t+t} \leftarrow \widehat{\boldsymbol{\theta}}^{\mathrm{GP}}_{h,t} - \eta \cdot \left(\nabla \widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}^{\mathrm{GP}}_{h,t}; \mathcal{D}) + \boldsymbol{w}_t \right)$$

  2. run n^2 iterations; then return the average $\widehat{\boldsymbol{\theta}}^{\mathrm{GP}}_h \leftarrow \frac{1}{n^2} \sum_{t=1}^{n^2} \widehat{\boldsymbol{\theta}}^{\mathrm{GP}}_{h,t}$
  3. ordering decision: $\widehat{q}(\boldsymbol{x}) = \left\langle \widehat{\boldsymbol{\theta}}^{\mathrm{GP}}_h, \boldsymbol{x} \right\rangle, \ \forall \boldsymbol{x}$



Gradient Perturbation

DP Algorithm: Gradient Perturbation (GP)

Theorem 1 (Data Security Gaurantee of GP)

Algorithm GP *is* (ε, δ)*-DP.*

Theorem 2 (Performance Guarantee of Algorithm GP)

With appropriate step size η*, kernel* K*, and bandwidth* h*, we have*

$$\sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathrm{GP};\mathbb{P}) \leq \mathcal{O}\left(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon} + \frac{1}{\sqrt{n}}\right)$$

- **Insight:** $\mathcal{R}(\mathrm{GP};\mathbb{P}) = \underbrace{\mathbb{E}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{GP}};\mathbb{P})\right] - \mathbb{E}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h;\mathbb{P})\right]}_{\text{security-induced error} \leq \mathcal{O}(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon})} + \underbrace{\mathbb{E}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h;\mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})}_{\text{statistical estimation error in non-DP case} \leq \mathcal{O}(\frac{1}{\sqrt{n}})}$

DP Algorithm: Objective Perturbation (OP)

- **Main idea**:
  1. Solve a regularized empirical risk minimization model on $\mathcal{D}$,

$$\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}} = \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \widehat{\mathcal{L}}_h^{\mathrm{OP}}(\boldsymbol{\theta};\mathcal{D}) := \frac{1}{n}\sum_{i=1}^{n} c_h(y_i - \langle \boldsymbol{\theta}, \boldsymbol{x}_i\rangle) + \lambda \|\boldsymbol{\theta}\|_2^2 + \frac{\boldsymbol{b}^\top \boldsymbol{\theta}}{n},$$

     with a perturbed objective function by Gaussian noise $\boldsymbol{b} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$, where $\sigma^2 \asymp \ln(1/\delta)/\varepsilon^2$.
  2. order quantity $\widehat{q}(\boldsymbol{x}) = \left\langle \widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}, \boldsymbol{x} \right\rangle, \forall \boldsymbol{x}$

DP Algorithm: Objective Perturbation (OP)

- **Main idea**:
  1. Solve a regularized empirical risk minimization model on $\mathcal{D}$,

$$\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}} = \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \widehat{\mathcal{L}}_h^{\mathrm{OP}}(\boldsymbol{\theta};\mathcal{D}) := \frac{1}{n}\sum_{i=1}^{n} c_h(y_i - \langle \boldsymbol{\theta}, \boldsymbol{x}_i\rangle) + \lambda \|\boldsymbol{\theta}\|_2^2 + \frac{\boldsymbol{b}^\top \boldsymbol{\theta}}{n},$$

     with a perturbed objective function by Gaussian noise $\boldsymbol{b} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$, where $\sigma^2 \asymp \ln(1/\delta)/\varepsilon^2$.
  2. order quantity $\widehat{q}(\boldsymbol{x}) = \left\langle \widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}, \boldsymbol{x} \right\rangle, \forall \boldsymbol{x}$

Theorem 3 (Data Security & Performance Guarantee of Algorithm OP)

Algorithm OP *is* (ε, δ)*-DP. Moreover, with appropriate kernel* K*, bandwidth* h*, regularizer* λ*, parameter* (ε, δ)*, we have*

$$\sup_{\mathbb{P}\in(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathrm{OP};\mathbb{P}) \le \mathcal{O}\left(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon} + \frac{1}{\sqrt{n}}\right)$$

The Cost of Data Security

- Theoretical lower bound:

$$\tilde{\Omega}\left(\frac{\sqrt{d}}{n\varepsilon}+\frac{1}{\sqrt{n}}\right) \leq \inf_{\pi\in\mathscr{F}_{\varepsilon,\delta}}\left\{\sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})}\mathcal{R}(\pi;\mathbb{P})\right\}$$

$$\leq \sup_{\mathbb{P}\in(\mathcal{X}\times\mathcal{Y})}\mathcal{R}(\text{GP or OP};\mathbb{P}) \leq \mathcal{O}\left(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}+\frac{1}{\sqrt{n}}\right)$$

- The cost of data security is $\mathcal{O}\left(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$

Simulations

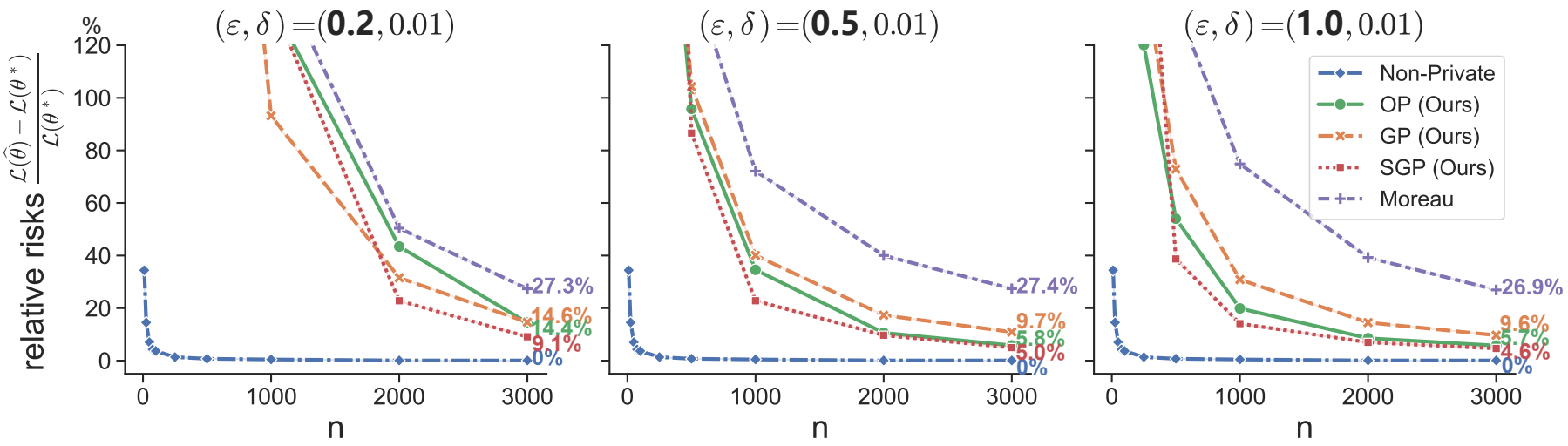


Figure: Relative risks of corresponding policies against true optimal risk in percentage. Demand follows $y = 10 + 5x_1 - 2x_2 + \epsilon(\boldsymbol{x})$, where $\epsilon(\boldsymbol{x}) \sim \mathcal{N}(x_1 - x_2 + |x_1 x_2|, 3^2)$

- Benchmarks:
  - "Non-Private": $\|\cdot\|_2^2$-regularized model, NV-ERM2 in Ban and Rudin (2019)
  - "Moreau": a private method by Bassily et al. (2019)
- Our methods are asymptotically optimal, and outperform existing methods

Data Security: the Curse of Being Small

1. **Small-sized** dataset:
   - "security for free" only when $n \gtrsim \frac{d \ln(1/\delta)}{\varepsilon^2}$
2. **Context-free** dataset:
   - protecting context through dropping context is financially impractical
3. **Limited number** of products:
   - security cost dilutes in # of products: $\mathcal{O}\left(\frac{1}{\sqrt{s}} \cdot \frac{\sqrt{d \ln(1/\delta)}}{n\varepsilon}\right)$

Tips for reducing security cost: Expand your $n \times d \times s$ dataset in all dimensions

Impacts on Upstream Supplier

In short, the upstream supplier will be worse off in all ways, as demand signaling process is further distorted by DP algos.

1. Supplier will obverse an increased variability in order quantity:

$$\Omega\left(\frac{\sigma^2}{n^2} - \frac{\lambda + h}{n}\right) \leq \Delta V(\widehat{q}_h^{\mathrm{OP}}(\boldsymbol{x}), \widehat{q}(\boldsymbol{x})) \leq \mathcal{O}\left(\frac{\sigma^2}{n^2}\right), \quad \forall \boldsymbol{x} \in \mathcal{X},$$

2. Supplier will be worse off, and gain a lower profit share (w.h.p.)

Conclusions and Takeaways

1. We considered a differentially private (DP) data-driven newsvendor problem for managing data security in supply chain.
2. We proposed several DP algorithms: Gradient Perturbation and Objective Perturbation; and we characterized the cost of data security $\mathcal{O}\left(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}\right)$.
3. We pinpointed "the Curse of Being Small" phenomenon in data security; expanding training dataset can mitigate the curse.
4. We identified negative external effects of DP on an upstream supplier.

Thanks!

chen1443@e.ntu.edu.sg (Chen Du 陈督)

Chen Du's homepage

Stay tuned!
Journal version
coming soon...